These tips are good to keep in mind. They are in no specific order. We add tips as we encounter common mistakes that students make.

Major tips

1. Do a literature study. The purpose of your literature study is to identify the missing knowledge (or sometimes even incorrect knowledge) which your thesis or article is going to contribute (or rectify). Be sure to present your literature survey with this in mind in your thesis as well; don't make it an unordered enumeration of what other people did.
2. Think of your research question(s). What is your thesis going to show? Although it is possible to start your work 'curiosity-driven' it is highly likely that you will get lost and wander in circles. It is much better to work focused towards an answer to a specific question, like whether there is a critical density of crowds for stop-and-go waves to occur, or whether the interest-rate swaps financial market is a self-organized critical system or not.
3. Meet with your supervisor regularly, and/or update him/her digitally in between. Not only does it keep your momentum, it also keeps the project fresh in the mind of your supervisor who probably has a lot more things to worry about.
4. Use Dropbox/Bitbucket/Google Drive or whatever other versioning control system which also backs up your files somewhere online. Nothing is more frustrating than either losing all your files (laptop crash) or accidentally introducing a bug in your code somewhere in the last 24 hours but you spend a week figuring out where.
5. Keep lab notes in something like Evernote or OneNote. Whenever you come up with new angles, research questions, results, possibly issues, possible conclusions, new hypotheses, free parameter values to still vary and test, have a meeting with your supervisor, etc., then make a note of it using a tool in which you can search back for them. Keep a separate TODO list which you continually update and which, by the end of your thesis, should be empty (or deferred to your next article as future work).
6. (*Computational Science*.) Keep in mind how you relate to computational modeling. You should either create a computational model to answer your research question (like an agent-based model of interest-rate swap contracts between banks), extend an existing model, or test/validate/invalidate an existing model.
    1. Although it is possibly for a thesis in Computational Science to not deal with any computational model at all, it is rare and more often a sign of a bad study setup, such as a blind and indiscriminate application of a textual classification algorithm to a Twitter dataset and simply showing the output. If your research question is, say, "Does homophily lead to stronger textual clustering" then you should first study this in an appropriate (networked) model. Then you could compare this to the clustering of real data, however keep in mind that your question was not "Is homophily the *only* possible cause of textual clustering" so in this case it may not even make sense to compare, unless you eliminate systematically most/all other potential causes.